

5.4 The Fundamental Theorem of Calculus

In this section we present the Fundamental Theorem of Calculus, which is the central theorem of integral calculus. It connects integration and differentiation, enabling us to compute integrals by using an antiderivative of the integrand function rather than by taking limits of Riemann sums as we did in Section 5.3. Leibniz and Newton exploited this relationship and started mathematical developments that fueled the scientific revolution for the next 200 years.

① Mean Value Theorem for Definite integrals

In the previous section we defined the average value of a continuous function over a closed interval $[a, b]$ to be the definite integral $\int_a^b f(x)dx$ divided by the length or width $b - a$ of the interval. The Mean Value Theorem for Definite Integrals asserts that this average value is always taken on at least once by the function f in the interval.

THEOREM 3 — The mean Value Theorem for Definite integrals

If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

Example 1 Show that if f is continuous on $[a, b]$, $a \neq b$, and if

$$\int_a^b f(x) dx = 0,$$

then $f(x) = 0$ at least once in $[a, b]$.

② Fundamental Theorem, part 1

It can be very difficult to compute definite integrals by taking the limit of Riemann sums. We now develop a powerful new method for evaluating definite integrals, based on using antiderivatives. This method establishes two basic relationships between

definite and indefinite integrals that together constitute a result called the “Fundamental Theorem of Calculus.” One part of this theorem will relate the idea of taking the limits of finite sums to obtain a definite integral, and the second part will provide a powerful method for evaluating definite integrals using antiderivatives. We begin by considering how to differentiate a certain type of function that is described as an integral.

THEOREM 4 —The Fundamental Theorem of Calculus, Part 1

If f is continuous on $[a, b]$, then

$$F(x) = \int_a^x f(t) \, dt$$

is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = f(x).$$

◆ The formula $F'(x) = f(x)$ states that $F(x)$ is an antiderivative of $f(x)$.

Example 2 Use the Fundamental Theorem to find $\frac{dy}{dx}$ if

$$(a) \, y = \int_a^x (t^3 + 1) \, dt;$$

$$(b) \, y = \int_x^5 3t \sin t \, dt;$$

$$(c) \, y = \int_1^x \cos t \, dt.$$

③ Fundamental Theorem, part 2

THEOREM 4 (Continued) — The Fundamental Theorem of Calculus, part 2

If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(t) \, dt = F(b) - F(a).$$

◆ To calculate the definite integral of f over an interval $[a, b]$ we need do only two things:

1. Find an antiderivative F of f , and

2. Calculate the number $F(b) - F(a)$, which is equal to $\int_a^b f(x) dx$.

Example 3 Calculate the following integral.

(a) $\int_0^\pi \cos x \, dx$.

(b) $\int_1^4 \left(\frac{3}{2} \sqrt{x} - \frac{4}{x^2} \right) dx$.

④ Total Area

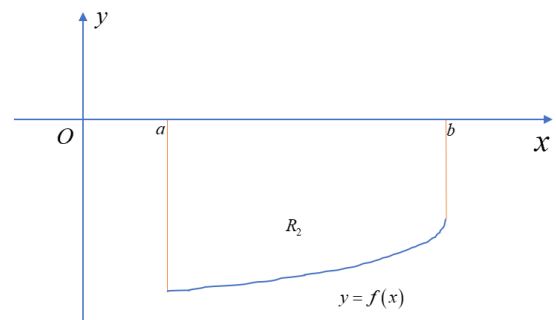
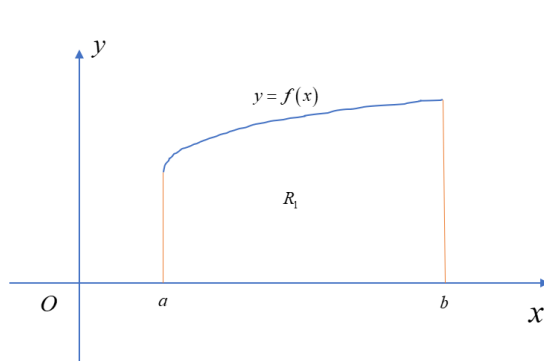
Area is always a nonnegative quantity. The Riemann sum approximations contain terms such as $f(c_k) \Delta x_k$ that give the area of a rectangle when $f(c_k)$ is positive. When $f(c_k)$ is negative, then the product $f(c_k) \Delta x_k$ is the negative of the rectangle's area. So, we have

(a) If $f(x) \geq 0$ is continuous over $[a, b]$, then

$$\text{Area of } R_1 = \int_a^b f(x) \, dx.$$

(b) If $f(x) < 0$ is continuous over $[a, b]$, then

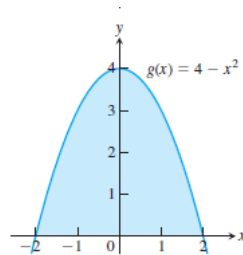
$$\text{Area of } R_2 = -\int_a^b f(x) \, dx = \left| \int_a^b f(x) \, dx \right|.$$



Note To compute the area of the region bounded by the graph of a function $y = f(x)$ and the x-axis when the function takes on both positive and negative values, we must be careful to break up the interval $[a, b]$ into subintervals on which the function doesn't

change sign. Otherwise, we might get cancellation between positive and negative signed areas, leading to an incorrect total. The correct total area is obtained by adding the absolute value of the definite integral over each subinterval where $f(x)$ does not change sign. The term “area” will be taken to mean this *total area*.

Example 4 Let $f(x) = -x^2 + 4$. Find the area between the graph and the x -axis over $[-2, 2]$.



Example 5 Find the area of the region between the x -axis and the graph of $f(x) = x^3 - x^2 - 2x$, $-1 \leq x \leq 2$.

